MIL-E-1/1017E
5 November 1975
SUPERSEDING
MIL-E-1/1017D
10 March 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5879

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, audio frequency, low noise, sharp cutoff

Outline	---	6-2 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	g1	nc	k	h	h	nc	g2	a	g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ecc2	Ec3	Pg2	Pp	Ehk	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	W	W	v	°C	ft
Maximum:	---	---	---	---	---	---	---	---	---	100	Note 2
Pentode:	6.9	330	---	165	330	---	0.15	1.4	100	---	---
Triode:	6.9	275	---	---	---	---	---	1.65	100	---	---
Minimum:											
Pentode:	5.7	---	---	---	---	---	---	---	---	---	---
Triode:	5.7	---	---	---	---	---	---	---	---	---	---
(See note 1)											
TEST CONDITIONS:	6.3	250	-3	100	---	0	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)	-		0.4	II	Ib	1.1	2.5	mAdc
1266	Total grid current	3	Rg1 = 0.5 Meg	0.4	II	Ic1	0	-1.0	µAdc
1306	Transconductance (1)	-		0.4	II	Sm	750	1250	µmhos
1306	Transconductance (2)	-	Ef = 5.7 V	0.4	II	Sm Ef	650	---	µmhos
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Rp = 10,000 ohms	---	---	Ep	---	10	mVac
1256	Electrode current (2) (anode)	-	Ec1 = -10 Vdc	1.5	S3	Ib	0	100	µAdc
1256	Electrode current (screen)	-		---	---	Ic2	0.2	0.6	mAdc
1301	Heater current	-		---	---	If	138	162	mA
1336	Heater-cathode leakage	-		---	---	Ihk	---	1.0	µAdc
---	Audio frequency noise	4	Ebb = Ecc2 = 250 Vdc; Rp = 0.1 Meg; Rg1 = 0.05 Meg; Rg2 = 0.39 Meg; Cg2 = 0.1 µF (min); Rk = 1,000 ohms; Ck = 50 µF (min); Ef = 6.3 Vac	4.0	S3	Eg1	---	50	µVrms
(E) 1239	Noise and microphonics, integrated	5	Ebb = Ecc2 = 250 Vdc; Rp = 0.1 Meg; Rg1 = 0.05 Meg; Rg2 = 0.39 Meg; Cg2 = 0.1 µF (min); Rk = 1,000 ohms; Ck = 50 µF (min); Ef = 6.3 Vac; t = 100 ms (min)	4.0	S3	EB	---	7.5	mV-sec
1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
2126	Glass strain	-		4.0	I	---	---	---	---
1121	Base strain	-		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP. LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: 1ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1501	Intermittent life	Group A; Ehk = 100 Vdc; Rg1 = 0.1 Meg	---	---	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours):									
1306	Transconductance (1)		---	---	---	---	Sm	650	---	µmhos

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NOTES:

1. Tie g2 and g3 to anode for triode connection.

2. See "Reduced pressure (altitude rating", and altitude, maximum peak voltage in the basic document.

3. This test to be performed at the conclusion of the holding period.

4. Pin 9 (g3) and the center-tap of the heater supply transformer shall be grounded. Pins 2 and 6 shall be grounded or bypassed to ground. Socket clips for the heater pins and all associated heater wiring shall be electrostatically shielded from other test socket circuitry.

   The response of the amplifier shall be within ±3 dB of the response at 400 Hz at all frequencies from 20 to 10,000 Hz. The indicator system which follows the amplifier shall respond to the true rms value of any waveform delivered by the amplifier.

   The amplifier may be calibrated by applying 1 mVac to the grid of a tube of average gain characteristics in the test socket and setting the amplifier gain for 1,000 μVrms indication. The noise reading is expressed in equivalent noise and hum voltage with respect to the grid of the tube under test. The noise limit shall apply to this reading.

(E) 5. The average of three consecutive taps is acceptable.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3004)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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